

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE ON
AgroVision

submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**AgroVision**” in partial fulfillment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Mr Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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I am also thankful to the faculty members of my department for providing academic support, resources, and a positive learning environment necessary for the successful completion of this project. Their teachings and constructive feedback greatly contributed to my knowledge of software development, deep learning, and system design.

I would also like to acknowledge my friends and peers whose suggestions, discussions, and motivation supported me during different stages of the project. Their ideas and encouragement helped improve various aspects of the implementation process.

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Finally, I express my sincere gratitude to everyone who directly or indirectly contributed to the successful completion of this project. This project has been a valuable learning experience and an important step in my academic and technical journey.

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Abstract

The AI-Powered Insect Detection and Pesticide Recommendation System is a smart agriculture solution designed to assist farmers in identifying crop pests quickly and accurately using artificial intelligence and computer vision technologies. The project follows a practical and user-friendly approach, focusing on real-time pest detection, sustainable farming practices, and easy accessibility for farmers through mobile and web platforms. The main objective of the system is to reduce crop damage, minimize excessive pesticide usage, and improve agricultural productivity through intelligent decision-making.

Unlike traditional pest identification methods that depend heavily on manual inspection and agricultural experts, the proposed system provides an automated and cost-effective solution capable of detecting insects directly from uploaded crop images. The system uses deep learning techniques, especially Convolutional Neural Networks (CNN), to classify insects based on their visual features and provide suitable pesticide recommendations according to the detected pest type and infestation level.

The application allows farmers to capture or upload crop images through a simple interface, after which the AI model processes the image and generates real-time results. The system displays the detected insect name, confidence level, recommended pesticides, preventive measures, and usage instructions. This enables farmers to take faster and more informed actions for crop protection.

Built using technologies such as Python, TensorFlow/PyTorch, OpenCV, and cloud-based databases, the project emphasizes accuracy, lightweight processing, and efficient performance. The implementation also focuses on scalability, ease of use, and environmental sustainability by promoting targeted pesticide application instead of excessive chemical use.

Overall, the project demonstrates how artificial intelligence and machine learning can be effectively integrated into agriculture to support smart farming, improve crop health, and enhance farmer productivity while protecting environmental resources.

Introduction

Modern agriculture faces major challenges due to crop damage caused by insects and pests, leading to reduced productivity and financial losses for farmers. Traditional pest identification methods mainly depend on manual inspection and expert consultation, which are often time-consuming, expensive, and inaccessible in rural areas. Incorrect pest identification frequently results in excessive pesticide usage, negatively affecting soil fertility, water quality, and environmental sustainability.

The AI-Powered Insect Detection and Pesticide Recommendation System was developed as a smart farming solution to address these challenges through artificial intelligence and computer vision technologies. The system focuses on providing real-time insect detection, accurate pest classification, and intelligent pesticide recommendations using uploaded crop images.

Unlike conventional methods, the proposed system automates pest identification using deep learning models, particularly Convolutional Neural Networks (CNN), enabling faster and more reliable decision-making for farmers. The platform aims to reduce unnecessary pesticide usage while improving crop protection and agricultural productivity.

The application allows farmers to upload crop images through a mobile or web interface, after which the AI model analyzes the image and displays the detected pest along with suitable pesticide recommendations and preventive measures.

Methodology

The methodology of the proposed system begins with collecting and preprocessing insect image datasets from agricultural sources. The images are resized, normalized, and enhanced to improve model accuracy. A Convolutional Neural Network (CNN) model is then trained using the processed dataset to identify and classify crop pests. Farmers can upload crop images through a mobile or web application, where the trained AI model performs real-time pest detection and classification. Based on the detected insect and infestation level, the system provides suitable pesticide recommendations and preventive measures to support effective and sustainable pest management.

Tools and Technologies

- Programming Language: Python
- Frontend Technologies: HTML, CSS, JavaScript
- Backend Framework: Flask / Django
- Machine Learning Frameworks: TensorFlow / PyTorch
- Image Processing Library: OpenCV
- Database: MySQL / Firebase
- AI Technique Used: Convolutional Neural Network (CNN)
- Cloud Services: Google Cloud / AWS
- Development Tools: Jupyter Notebook, VS Code
- Operating System: Windows / Linux

Implementation

The implementation of the AI-Powered Insect Detection and Pesticide Recommendation System begins with collecting and preparing insect image datasets from agricultural sources. The images are preprocessed using image enhancement techniques such as resizing, normalization, and augmentation to improve the performance and accuracy of the AI model.

A Convolutional Neural Network (CNN) model is developed and trained using frameworks such as TensorFlow or PyTorch for insect classification. The trained model is integrated into a backend system developed using Python and Flask/Django. Image processing and pest detection operations are performed using OpenCV.

A mobile/web-based user interface is created to allow farmers to upload crop images easily. Once an image is uploaded, the system processes it in real time, identifies the insect type, and generates pesticide recommendations based on the prediction results. The detection data, pesticide details, and user records are stored in a database such as MySQL or Firebase for future analysis and monitoring.

The final system is tested for accuracy, responsiveness, and usability to ensure reliable performance under different agricultural conditions.

Conclusion and Future Scope

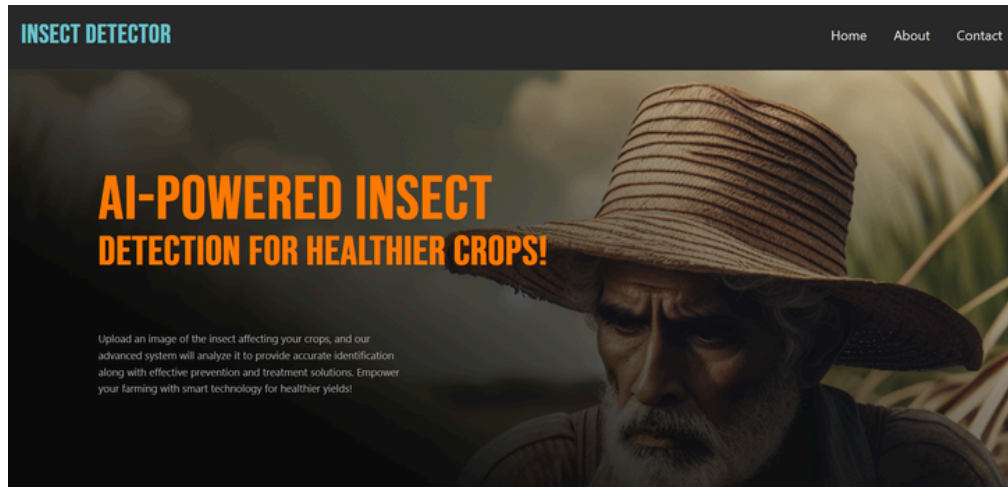
The AI-Powered Insect Detection and Pesticide Recommendation System provides an effective and intelligent solution for modern agricultural challenges. By combining artificial intelligence, machine learning, and computer vision, the system can accurately identify crop pests and provide real-time pesticide recommendations. This helps farmers reduce crop damage, improve productivity, and make faster decisions for pest management.

The project also promotes sustainable farming practices by minimizing excessive pesticide usage and reducing environmental impact. Through targeted pesticide application, the system supports better soil health, lower farming costs, and safer agricultural methods. The user-friendly mobile and web interface further ensures accessibility for farmers, including those in rural areas.

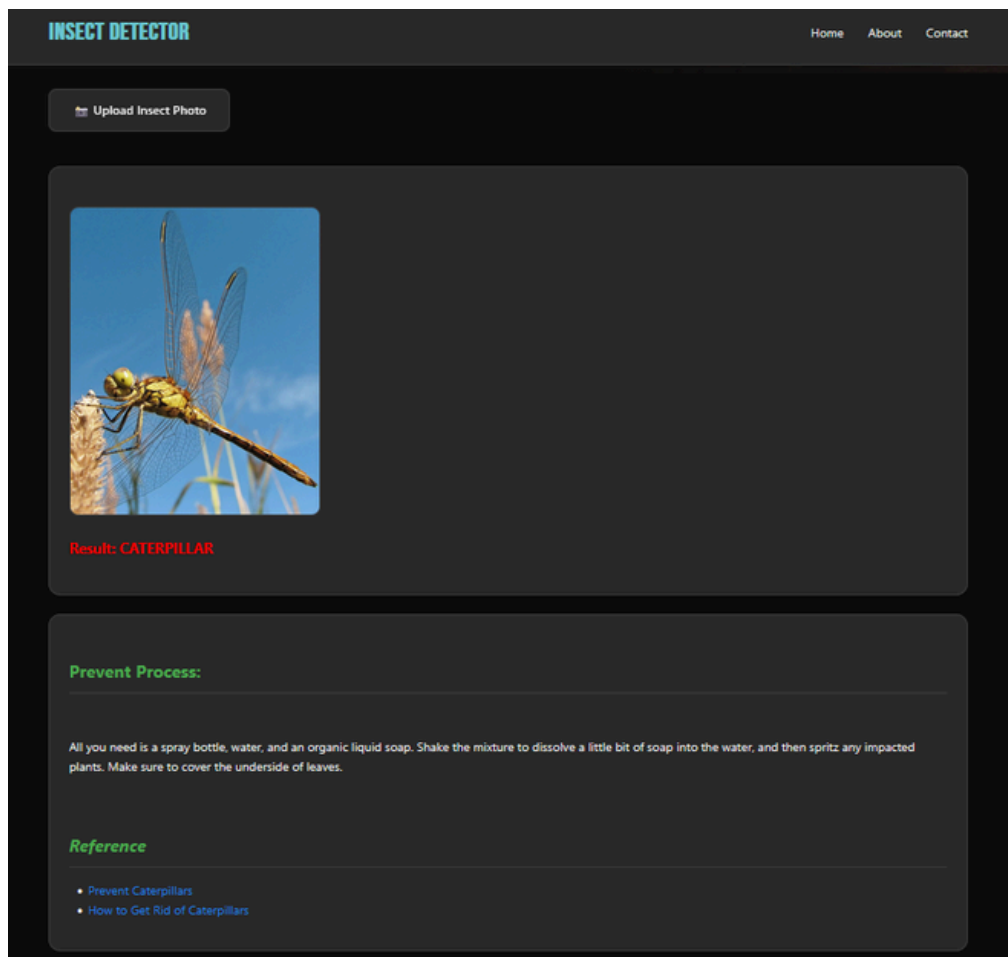
Overall, the project demonstrates the practical application of AI in agriculture and highlights the potential of smart farming technologies in improving crop protection and agricultural efficiency.

Result and Screenshotst

- Home page



- working



FutureScope

In the future, the system can be enhanced by integrating IoT sensors and drone technology for automatic field monitoring and large-scale pest detection. The platform can also be expanded to detect plant diseases, nutrient deficiencies, and crop health conditions alongside insect identification.

Additional improvements may include multilingual support, voice assistance for farmers, and integration with weather forecasting systems to predict pest outbreaks in advance. The project can further evolve into a complete AI-based smart farming platform using cloud computing, mobile technologies, and real-time agricultural analytics.

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